

ASSIGNMENT #1

Course: CHE 433. PROCESS SIMULATION WITH ASPEN PLUS
Term: Fall 2019
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Your name: Mariana St Peter
Due Date: Friday September 06 by 5:00pm in my office

Number of points.....100
Your score.....95.....

Note: All problems are worth 100 points. Your final score will be the average value

Problem 1: Review of Macroscopic Balances and Numerical Methods Used in Commercial Process Simulators.

Knowns

$$L \text{ (length)} = 36\text{ft} = 10.9728\text{m}$$

$$K \text{ (dirt factor)} = 0.0077\text{mm} = 0.0077 \times 10^{-3}\text{m}$$

$$Re \text{ (Reynolds number)} = 318,000$$

$$D \text{ (inside diameter)} = 1.049 \text{ in} = 0.0266446 \text{ m}$$

$$T1 \text{ (Inlet temperature)} = 70^\circ\text{F} = 294.261\text{K}$$

$$\text{Outlet temperature (T2)} = 170^\circ\text{F} = 349.817\text{K}$$

$$P1 \text{ (Inlet pressure)} = 20 \text{ psig} = 239220 \text{ Pa}$$

$$\rho \text{ (Density of air at } 70^\circ\text{F and } 20 \text{ psig)} = 0.177 \text{ lb/ft}^3 = 2.84 \text{ Kg/m}^3$$

$$\mu \text{ (Viscosity of air at } 70^\circ\text{F and } 20 \text{ psig)} = 0.3797 \times 10^{-7} \text{ lb-s/ft}^2 = 1.818 \times 10^{-5} \text{ Pa}\cdot\text{s}$$

$$M \text{ (Molecular weight of air)} = 28.97 \text{ g/mol} = 28.97 \times 10^{-3} \text{ kg/mol}$$

$$\text{Colebrook Equation: } \frac{1}{\sqrt{f}} = -1.737 \ln \left(\frac{k}{3.707D} + \frac{1.256}{Re\sqrt{f}} \right)$$

Objectives:

Calculate the friction factor using the Colebrook equation and the following methods

1. Newtons Method
2. Secant Method
3. Direct Substitution
4. Broyden Method
5. Wegstein Acceleration Method
6. Calculate the pressure drop and rate of heat transfer to the air.

Using the Colebrook equation in the format:

$$g(f) = \frac{1}{\sqrt{f}} + 1.737 \ln \left(\frac{k}{3.707D} + \frac{1.256}{Re\sqrt{f}} \right) = 0$$

Its derivative will be:

$$g(f) = \frac{1}{\sqrt{f}} + 1.737 \ln\left(\frac{k}{3.707D} + \frac{1.256}{Re\sqrt{f}}\right) = 0$$

Secant Formula:

$$x^{k+1} = x^k - \frac{g(x^k)}{g(x^k) - g(x^{k-1})} (x^k - x^{k-1})$$

(see attachment for calculations)

Solution: $f = 0.00420143$

Part 3) Direct Substitution

$$\varphi = \frac{1}{\sqrt{f}}$$

$$g(\varphi) = -1.737 \ln\left(\frac{k}{3.707D} + \frac{1.256}{Re} * \varphi\right) = 0$$

(see attachment for calculations)

Solution: $f = 0.00420143$

Part 4) Broyden Method

$$g(x) = \frac{1}{\sqrt{f}} + 1.737 \ln\left(\frac{k}{3.707D} + \frac{1.256}{Re\sqrt{f}}\right) = 0$$

$$g'(x) = \frac{-1}{2} f^{-3/2} + 1.737 \left(\frac{1}{\frac{k}{3.707D} + \frac{1.256}{Re\sqrt{f}}} \right) \left(\frac{-1.256}{2Re} \right) f^{-3/2}$$

$$x_1 = x_0 - (g(x_0) * g'(x_0))$$

(see attachment for calculations)

Solution: $f = 0.00420143$

Part 5) Wegstein Acceleration

$$-G^2 \ln \left(x \frac{T_1}{T_2} \right) + \frac{M_w P_1^2}{2RT_a} (x^2 - 1) + G^2 \frac{2Lf}{D} = 0$$

$$T_a = \frac{T_1 + T_2}{2}$$

$$x = P_2 / P_1$$

After adding the values for the G^2 equation, a random number was chosen for P_2 . Goal seek was used to set the equation to zero, and a $P_2 = 145979.69$ Pa

The pressure drop was the 93240.31 Pa = 13.523 Psia.

Calculating the heat

First the density was calculated for the inlet/ outlet by using $\rho = \frac{P}{T \cdot R} \cdot M$

Then the inlet/ outlet velocity was calculated using $G = \rho \cdot v$

(values are displayed on Excel)

Next, the formula:

$$Q = G \cdot \frac{\pi \cdot d^2}{4} \cdot \left(C_p \cdot (T_2 - T_1) + \frac{1}{2} (v_2^2 - v_1^2) \right)$$

was used. The velocity was divided by 1000 to convert the units to the proper format.

After solving for this equation, a heat flow rate of 7.7235 kJ/s was found. Which converting to Btu/h is 26353.6759 .

L =	10.9728
k =	0.0000077
Rc =	318000
d =	0.0266446

Newton's Method				
$x(k)$	$f(x(k))$	$f'(x(k))$	$x(k+1)$	
0.001	16.85306	-16346.1	0.002031	
0.002031012	7.067081	-5688.93	0.003273	
0.00327363	2.149764	-2794.54	0.004042	
0.004042434	0.315546	-2040.66	0.004197	
0.00419692	0.008981	-1923.78	0.004201	
0.004201427	6.952E-06	-1926.7	0.004201	
0.004201431	4.45E-12	-1926.7	0.004201	

Secant's Method				
$x(k-1)$	$f(x(k-1))$	$x(k)$	$f(x(k))$	$x(k+1)$
0.001	16.85306	0.002	7.245539	0.002754
0.002	7.245539	0.003273	1.386034	0.004036
0.003273	1.386034	0.004042	0.329098	0.004183
0.004042	0.329098	0.004197	0.035615	0.004201
0.004197	0.035615	0.004201	0.001047	0.004201
0.004201	0.001047	0.004201	3.42E-06	0.004201
0.004201	3.42E-06	0.004201	3.3E-10	0.004201
0.004201	3.3E-10	0.004201	0	0.004201

Direct Substitution			
$P(x)$	$f'(x)$	f	
0.1	16.42210298	100.00000000	
16.42210298	15.37927062	0.003708023	
15.37927062	15.43010147	0.004227937	
15.43010147	15.42758903	0.00420127	
15.42758903	15.42771313	0.004201485	
15.42771313	15.427707	0.004201428	
15.427707	15.427703	0.004201431	

Broyden's Method							
x	$f(x)$	$f'(x)$	$x - f(x)$	$1/(1 - f'(x))$	dx	df	
0.009	0.013394	2.261068	-0.00439431	-0.79298			
0.005515409	0.00636	1.755886	-0.00084473	-1.32293	0.003485	0.007034	
0.004397881	0.004505	1.561334	-0.00010659	-1.78147	0.001118	0.001356	
0.004207817	0.004211	1.525902	-3.3549E-06	-1.9015	0.00019	0.000093	
0.00420438	0.004201	1.524699	-3.8361E-09	-1.90585	6.38E-06	9.73E-06	
0.004201431	0.004201	1.524698	-5.0376E-15	-1.90586	7.31E-09	1.11E-08	
0.004201431	0.004201	1.524698	0	-1.90586	9.6E-15	1.46E-14	

Wegestein Acceleration						
$P(x(k-1))$	$P(x(k))$	$f(P(x(k-1)))$	$f(P(x(k)))$	s	q	$P(x(k+1))$
0.001	0.002	16.43079	16.43071	-0.088	0	16.43071
0.002	16.43075	16.43071	15.37886	-0.06402	0	0.00370442
16.43075	15.378857	15.37886	15.43012	-0.04874	0.080880252	0.00428164
15.378857	15.42976	15.43012	15.42779	-0.04943	0.060172452	0.004202374
15.42976	15.427683	15.42779	15.42771	-0.0494	0.046472568	0.004201444
15.427683	15.427707	15.42771	15.42771	-0.0494	0.047103395	0.004201431
15.427707	15.42771	15.42771	15.42771	-0.0494	0.047103395	0.004201431

P1=	239220
T1=	294.261
T2=	349.817
Ta=	322.039
G=	216.021258
Mw=	0.02897
L=	10.9728
d=	0.0266446
P2 initial guess =	150000
f=	0.004201431
R=	8.314
Re=	318000
Miu	0.0000181
Cp (70F) kJ/(Kmol*k)	29.1400
Cp (170F) kJ/(Kmol*k)	29.26
Cp average kJ/(Kmol*k)	29.2
Cp (ta)kJ/Kg*K	1.00793925
Inlet density kg/m3=	2.83271743
outlet density kg/m3	1.4541
inlet velocity, m/s	76.2594
outlet velocity, m/s	148.5616
P1 - P2	93240.3100
G Equation After Goal Seek	0.0000

$$-G^2 \ln \left(x \frac{T_1}{T_2} \right) + \frac{M_w P^2}{2RT_g} (x^2 - 1) + G^2 \frac{2Lf}{D} = 0$$

$$P2 \text{ after goal seek} = 145979.7 \text{ Pa}$$

$$Q = G * \frac{\pi * d^2}{4} * \left(Cp * (T2 - T1) + \frac{1}{2} (v2^2 - v1^2) \right)$$

$$Q \text{ heat flow rate KJ/s} = 7.723544$$

$$Q \text{ heat flow rate Btu/h} = 26353.68$$